

Slide 1: Team Details

Track: Healthcare — Good Health and Well-being

Problem Statement: P10 — From One Empty Shelf to a Regional Shortage

Solution Title: SupplySense — Early Warning System for Medicine Shortages

(alt titles if you want options: "ShelfWatch," "MedPulse," "Sentinel Stock" — I'd go with SupplySense, it directly names what the system does)

[Team name, members, roles — as per template]

Slide 2: Solution

One-line statement:

SupplySense connects village, block, and district health facilities on one platform to catch medicine shortages early — before an empty shelf becomes a regional crisis.

The Problem:

- Facilities track stock in isolation — a shortage is invisible to anyone outside that facility until it's already critical
- No one connects the dots when *multiple* facilities run low on the *same* medicine at the *same* time
- Surplus stock sits unused at one facility while a nearby facility stocks out

Our Solution:

- A shared dashboard across facility, block, and state levels — real-time visibility of stock, consumption, and replenishment
- An algorithm that flags **two kinds of alerts**: local stockout risk *and* regional shortage patterns (many facilities trending toward the same stockout)
- A map-based view showing medicine availability by geography, with smart redistribution suggestions between nearby facilities
- Every recommendation is human-approved — the system informs decisions, it doesn't act on its own

Impact: Faster response, fewer stockouts, less wastage, better use of existing stock.

Slide 3: Technical Implementation

System Structure:

- Facilities organized in a hierarchy: Village Health Center → Block Hospital → District Warehouse → State
- Two data flows: supply moves **downstream** (replenishment), shortage signals move **upstream** (alerts)

How it works:

1. **Facility level:** Admins log periodic stock updates (received/dispensed); system computes days-of-stock-remaining per medicine
 2. **Detection:** Algorithm flags status (Green/Yellow/Red/Critical) based on stock trend and consumption rate, with an uncertainty range — not a false-precision number
 3. **Regional signal:** If multiple facilities in a cluster trend toward the same shortage simultaneously, system raises a *regional* alert, distinct from individual facility alerts — this is the early warning for a systemic disruption
 4. **Redistribution matching:** System scans nearby facilities for surplus, ranks matches by distance and urgency, and suggests transfers
 5. **Human-in-the-loop:** District/block managers review and approve every replenishment and redistribution — the system recommends, humans act
 6. **Map view:** Select any medicine → see live availability across all facilities geographically
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Slide 4: Feasibility

Tech Stack:

- Frontend: React | Backend: FastAPI | Map: Mapbox
- Rule-based detection algorithm (rolling consumption average + threshold logic) — explainable and lightweight, no black-box ML required at pilot stage

Why it's buildable now:

- Core logic (thresholds, trends, geographic matching) doesn't require complex infrastructure — deployable on modest hardware, works even with basic internet connectivity
- Simulated/public health supply data used to demonstrate and validate the approach before real deployment

Rollout Plan:

- **Phase 1:** Pilot in a single district — a handful of facilities, one supply chain
- **Phase 2:** Expand to neighboring blocks/districts once workflows are validated
- **Phase 3:** As real usage data accumulates, refine the detection algorithm for better accuracy and lead time

Slide 5: Business Strategy

Model: Not a consumer product – built for government health systems and public health programs.

Monetization:

- Subscription/licensing model for state health departments (aligned with existing government procurement channels)
- Pilot funding via public health NGOs or donor programs, with state ownership as the long-term goal

Why it pays for itself (ROI):

- Reduces emergency/expedited procurement costs caused by last-minute stockouts
- Cuts medicine wastage from expired surplus by enabling redistribution
- Reduces manual coordination effort for health administrators

Go-to-Market:

- Engage district/state health officials directly with pilot results
- Demonstrate impact on a small scale before scaling state-wide
- Partner with existing public health infrastructure rather than replacing it